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Senior IT Consultant

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Professional Experience with Reference Projects

Requirements Engineer, Technical Product Owner

Timeframe: Apr 2022 -

Employer: RISE (Research Industrial Systems Engineering) GmbH (Vienna, Austria)

Client: Gematik (Digital Healthcare Agency of Germany)

Requirements engineering of the following RISE products (all being central or edge components of the German Healthcare Telematics Infrastructure):

- Central IDP (Identity Provider)
- Federation Master
- KIM 1.5 (Encrypted communications platform of the telematics infrastructure)
- HSKVS (Management software for the 'RISE High-Speed-Connector' – a high-performance multi-tenant security hardware for accessing the telematics infrastructure, used by hospitals, insurance companies and IaaS providers)
- TiaaS (Telematics Infrastructure as a Service, from 2024 branded as RISE TI-Gateway)

Senior Lecturer

Timeframe: Sep 2021 – Feb 2022

Client: IBS International Business School, Budapest/Vienna/Remote

Teaching software development courses in IBS's 'Masters in IT for Business Data Analytics' programme (a joint degree programme with the University of Buckingham)

CTO, Lead Architect

Timeframe: Jan 2019 – Jun 2020

Employer: Enernova GmbH (Luzern, Switzerland)

Enernova GmbH is a Swiss company founded by senior managers of Siemens Building Technologies. As such, its main focus is developing software products (mainly in the area of business process management / automation) for Siemens Building Technologies. My roles included the organization of the company's new (partially remote) development team including the design of software development rules, guidelines and workflows. In addition, I carried out technical product design and development on the following products:

- CEDB: Commissioning and Engineering DataBase: A business application for managing the data and logistics processes required for the deployment and testing of Siemens Desigo based building automation solutions (CUBA, Spring, Spring Integration, Activiti, Vaadin, PostgreSQL).
- Simcaat and QuickCheckTool: Web based mobile tools (PWAs) for carrying out technical check-ups on buildings in order to provide building owners with a price quote for new energy optimization solutions based on Siemens automation products (Angular, PWA, Spring, ReST,

PostgreSQL, Jira)

- MET: A Jira based tool suite integrated with other Enernova and Siemens products for managing problem ticket workflows related to the deployment and testing of building automation solutions, including an offline web client (PWA) for Jira that replicates its configurable data entry forms and workflows, and lets engineers record tickets in environments where no stable network connection is present (i.e. in basements where many technical devices of crucial importance are located).

Freelance IT Professional (Architect / Consultant / Developer / Instructor)

Timeframe: Aug 2015 – Current

Client: Contractor for various Europe and US based clients (in the Engineering, Healthcare and Academic sectors)

Projects (selected examples):

Google Netherlands BV, Allianz Technology GmbH

Google Netherlands develops the 'G4' toolkit, originally produced by Cornerstone BV, which is a set of analysis and conversion tools for automatically transforming legacy mainframe applications (written in legacy languages, e.g. COBOL or PL/I) to state-of-the-art technologies. In this specific projekt, where I worked as a senior consultant, we adapted the G4 engine to perform the conversion of Allianz's core business services ('ABS') from IBM PL/I to a cloud-native Java Spring application, thereby further developing G4's conversion logic as well as its Java based emulation layer (that serves as a drop-in replacement for several PL/I API-s, data types and language features) according to the requirements posed by Allianz's code base.

Braining Hub (Budapest, Hungary):

Lead instructor and designer of courses on Java (J2SE, J2EE, Spring), Angular, Node.JS, Laravel, Software testing and development methodology. Course sessions include internal bootcamp groups as well as externally held courses aimed towards mid-level to senior IT professionals (clients include: T-Mobile Hungary, Bosch Hungary, Intrum Justitia Zrt). Some courses included additional consultancy services for refactoring existing products of the company based on the new technological stack being taught.

VetaTek LLC (San Diego, USA):

Architectural design and development of MetaBase, a metadata based business application development platform (NodeJS, Angular 5, MSSQL, Azure Cloud, Azure Application Insights, TFS Continuous Integration)

Institute of Ethnography of the Hungarian Academy of Sciences (Budapest, Hungary):

Participation in computer-aided music analysis and ethnomusicological archive digitization projects (ongoing work, about to be published).

WSCAD GmbH (Bergkirchen, Germany):

Development of a single sign on (SSO) solution based on WSO2 Identity Server extended by custom built plugins (J2EE)

Development of the Building AR mobile application for Android, which is capable of creating an editable floor plan of a building by recording its dimensions in an Augmented Reality environment

InnoTrain Kft (Budapest, Hungary):

Design and development of new version of the Online Trainer platform, which provides an online „virtual personal trainer“ service by using computer generated training plans based on digitally measured and monitored health data. (Java EE, Spring, GraphQL, AspectJ, Maven, Jenkins, NodeJS, PostgreSQL, MongoDB, Android)

IDBS E-Workbook and EMC Documentum® integration layer

Timeframe: Sep 2015 – Nov 2015

Employer: DACHS Computing and Biosciences GmbH (Budapest office)

Client: Novartis AG, Basel, Switzerland

Architectural design and development of a synchronization middleware between IDBS E-workbook, where medical research data is generated, to EMC Documentum®, where research documentation goes through the review and approval process. (Java EE, Spring, Oracle SQL DB, WebLogic)

VirtualEDI (EMC Documentum® Client)

Timeframe: Mar 2015 – Sep 2015

Employer: DACHS Computing and Biosciences GmbH (Budapest office)

Client: Novartis AG, Basel, Switzerland

A client application for an EMC® Documentum 2 system, developed specially for Novartis Pharma GmbH for worldwide use, which manages documentation and handles document lifecycles for medical research experiments.

EcoLane (fleet management and route planning)

Timeframe: Oct. 2014 – Mar 2015

Employer: ITWare Kft.

Client: EcoLane LLC, Espoo, Finland

EcoLane develops software which is mainly used by ambulance dispatchers in the United States. The present development involved re-writing their present one-tier PHP application to a three-tier Java EE system with a Ruby on Rails front-end.

MVNE (Mobile Virtual Network Enabler)-capable CRM development and GSM systems integration

Timeframe: Feb 2014 – Oct 2014
Employer: ITWare Kft., EuroMACC Kft.
Client: Ventocom GmbH, Vienna, Austria

Ventocom is a virtual mobile network enabler (MVNE) in Austria, founded by the past leaders of Orange Telecom. Our collaboration with them involved the development of an enterprise CRM system to allow the creation and management of any number of virtual mobile network operators (MVNO-s). I was in charge of designing and developing the integration with T-mobile's Home Location Register (HLR) and Mobile Network Portability (MNP) systems, as well as the subsystem for processing and generating SMS and USSD messages, in close cooperation with T-Mobile Austria's engineers. As per today, this system serves several virtual mobile operators including Allianz-SIM (<https://www.allianz-sim.at/>) and Hofer Telekom (<https://www.hot.at/>)

TransRound Android application translation framework

Timeframe: Oct 2013 – Mar 2015
Employer: ITWare Kft.
Client: TransRound Kft.

This is a framework which is capable of altering the UI of Android applications as to allow end users to translate and localize applications from within the application itself. The system is built by taking advantage of the AspectJ technology. I worked on the development of the core AspectJ component, the mobile application, and designed the backend and web services layer (collaborative translation facilities, machine translation integration, license management, etc.)

Mobile and backend development with Arteries Studio Kft.

Timeframe: Jan 2013 – Oct 2013
Employer: Arteries Studio Kft.

Cardy Android Application: Software which allows controlling Cardy ECG devices from mobile phones and tablets, displays measurement results, and synchronizes measurement data with a cloud back-end.

Fish and Chips E-Magazine Reader: This is an E-Magazine reader application for Android and iOS, where I was involved with developing the applications' backend.

ScooterMonitor.nl Android Application and Backend: This application enables users to track vehicles equipped with GPS, and it also provides users with diagnostic information acquired via the CAN bus protocol. I was one of the developers of the Android application and I designed its REST API protocol.

E-fit and Online Trainer (OT) Services: The first (proof-of-concept) version of the applications which later turned to InnoTrain's new product (see above).

Freelance projects (a selected few examples)

Timeframe: 2008-2012

FISEC International Student's Olympic Games: Online registration and result management system, where schools could register their competitors and teams, and referees could store their results, which were then populated as a searchable database.

Magnalex Kft: Interactive distance education portal based on Moodle. The Moodle base system has been extended with custom developed plug-ins, and with an enrollment management system, which reads and processes tuition fee payments from exported online banking account statements. These extensions also enable the management of company licenses, where users can enroll into courses if they access the site from within a given IP range. Students can download video based training material from the portal, which have their own installer, and which also require an online activation, protecting them against unauthorized use.

Oktatovideok.hu: Copy protection system using hardware identification and online activation to protect IT training materials published by this company.

Central Institute for Physics, Budapest (KFKI): Data processing system for ordering and evaluating electron cannon experiments.

Areas of Expertise

Courses and Certifications

Introduction to Accessibility and Inclusive Design (Coursera/University of Illinois at Urbana-Champaign)	2023
Design Thinking: Insights to Inspiration (Coursera/University of Virginia)	2023
Google SEO Fundamentals (Coursera/UC Davis)	2023
Certified Professional in Requirements Engineering - Foundation (IREB)	2023
Certified Scrum Product Owner (Scrum Alliance)	2022
Neuromatch Academy - Computational Neuroscience	2021
Machine Learning (Coursera/Stanford, Andrew Ng)	2017
Embedded Systems (EDX/UTAustin, Valvano/Yerraballi)	2016
R Programming (Coursera/Johns Hopkins, Peng/Leek/Caffo)	2015
Software Security (Coursera/UMaryland, Michael Hicks)	2015

Key Soft Skills

An enthusiastic curiosity
An entrepreneurial mindset

The ability to inspire and to motivate others

A drive to think about interesting solutions to problems, human or technical

The openness to learn, teach and to listen

Programming Languages, Technologies and Frameworks

Java/Kotlin	Java SE Android development Spring Framework EJB CUBA Platform Struts
Swift	iOS development (SwiftUI + UIKit)
Ruby	Ruby on Rails
Python	Django PyTorch Scikit-Learn
Javascript/Typescript	Node.JS Angular React Vue.JS jQuery Next.js
PHP	Symfony Laravel phpUnit
C	
C++	
C#	.NET (Desktop+Web)
R	
Matlab/Octave	

Databases

PostgreSQL (with mid-level pl-pgSQL experience)
Oracle (with basic plSQL experience)
MySQL/MariaDB
SQLite
MongoDB
Neo4J
Redis
Firebase (Cloud FireStore)
CouchBase

Miscellaneous IT skills

Linux system administration
Knowledge of GSM specific protocols (especially SMPP)
Knowledge of the FIX (Financial Information Exchange) protocol
Issue trackers: Jira (with experience in plugin development), BitBucket, Confluence (administration + extension development)
Redmine, Mantis, Bugzilla (administration)
UML
Experience with Agile methodologies
Version trackers: GIT, SVN, Mercurial, CVS
Testing tools: JMeter, SoapUI, LoadUI, Selenium, Cucumber
Build tools: Ant, Maven, Gradle, Webpack, Gulp
Machine learning (intermediate level, R, Python)
Containerisation (Docker, Kubernetes, OpenShift, Istio)
Cloud providers (AWS, Azure, GCP, IBM Cloud)

Education

Timeframe: 2025-

Institution: Johannes Kepler University, Linz

Degree: MSc in Artificial Intelligence

Timeframe: 2019-2023

Institution: Budapest University of Technology

Degree: MSc in Computational and Cognitive Neuroscience

Timeframe: 2017-2020

Institution: University of Pécs

Degree: MA in Cultural Anthropology

Timeframe: 2014-2020

Institution: Vilmos Apor College

Degree: BA in Choral Conducting and Church Music

Language Skills

Hungarian Mother tongue

English Advanced (C2, IELTS overall score: 8.5/9)

German	Advanced (C1, Goethe Institut + ÖSD)
Dutch	Beginner
French	Intermediate

Miscellaneous skills

Driving license	Category „B“
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